# ORF307 Optimization

## 2025 Fall ORF397 Final Exam

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### Instructions

- Exam files: the exam problems (as a PDF) are available to download at Gradescope. Starter Python code and Notebook are available on Canvas → Assignments → Final Exam.
- Date and time: From December 13, 7:00 AM to December 14, 7:00 PM.
- Write your full name and honor code at the top of your solutions.
- Write your answers clearly and legibly.

### Exam Rules

- You are allowed to use all course materials on the final exam (lecture notes, books, precept materials, code, and homeworks). You cannot use internet to search for answers directly related to the problem in the exam.
- Strictly Prohibited: The use of any AI-assisted tool (e.g., ChatGPT, Gemini, Claude, etc.) is strictly prohibited.
- You have to justify all your answers to get full credit. If you use code from the course materials, you have to explain what each step means.
- The output and results you present must be verifiable from your included code.
- You cannot communicate with anyone during the exam.
- No late submissions allowed. Make sure your submission goes through on time. You can resubmit as many times as you like until your time expires.
- The exam must be submitted electronically as a single PDF on Gradescope before the deadline December 14, 7:00 PM.

**Note**: Problem 1 and Problem 2 are related and must be solved in sequence. Problem 3 is independent of the other.

1. *Problem 1 (30 points). Rural Lifeline Logistics - Optimal Redistribution*

Rural Lifeline Logistics operates six (6) regional centers and needs to efficiently **redistribute** a life-saving medicine among all centers. For each center, the company knows the units that are available ($b_i$) and the required demand ($d_i$). The corresponding data is shown in Table 1.

| **Center** $i$ | **Units Available** $b_i$ | **Units Required** $d_i$ |
|---|---|---|
| 1 | 10 | 15 |
| 2 | 20 | 10 |
| 3 | 5 | 10 |
| 4 | 15 | 20 |
| 5 | 10 | 5 |
| 6 | 15 | 15 |

**Table 1**

The transportation network and corresponding unit costs are summarized in Table 2. For example, transporting one unit of medicine from center 1 to center 2 costs \$2.5. The main goal of the company is to determine the optimal number of units to transport between centers to satisfy all required demand while minimizing the total transportation cost.

| **Arc** $(i, j)$ | **Cost (\$)** $c_{ij}$ | **Arc** $(i, j)$ | **Cost (\$)** $c_{ij}$ |
|---|---|---|---|
| (1, 2) | 2.50 | (3, 5) | 2.00 |
| (2, 4) | 1.80 | (4, 5) | 0.90 |
| (2, 6) | 3.10 | (5, 1) | 4.00 |
| (3, 4) | 1.20 | (6, 3) | 1.50 |

**Table 2**

(a) (5 points) For each center, compute the net flow requirement $R_i = b_i - d_i$. Based on the values you find clearly categorize each center as a supply node, a demand node, or transhipment node.

(b) (5 points) Formulate the company's problem as a Minimum Cost Flow LP problem. Clearly state and explain the decision variables, objective, and flow conservation constraints.

(c) (5 points) Use `CVXPY` to solve the LP problem. Report the optimal total cost ($Z^*$) and list the optimal flows ($x_{ij}$) for all arcs.

(d) (5 points) Explain why the optimal solution consists of integer values (no need to prove it).

(e) (5 points) Suppose that due to a car accident, we can only transport at most 4 units of medicine along arc $(2, 4)$ (perhaps due to closure of some lanes). Modify your original `CVXPY` code from part (c) to include this new constraint.

   i. Report the new optimal total cost $Z^{**}$.

   ii. List the new set of optimal flows.

   iii. Explain the main reason the total cost changed.

(f) (5 points) Define the Dual of the Minimum Cost Flow LP problem in part (b), and solve it using `CVXPY`.

   i. Report the dual optimal value $\Omega^*$

   ii. List the values of the optimal dual variables.

   iii. Briefly discuss their economic interpretation.

2. *Problem 2 (35 points). Demand Prediction, Multi-Objective Least Squares, and Re-Optimization.*

Rural Lifeline Logistics has realized that the "Units Required" ($\boldsymbol{d}$), used in the redistribution model in Problem 1, are a critical source of uncertainty and may change in the future. To create a more robust logistics plan, the company has decided to use historical data to develop a predictive linear model for this quantity. The goal of this problem is to find a set of predictive coefficients ($\boldsymbol{\beta}$) and then use the resulting forecasts to re-optimize the medicine's transportation network.

The company uses a linear model to predict the required units of medicine for center $i$:

$$\hat{Y}_i \approx \beta_0 + \beta_1 X_{i,1} + \beta_2 X_{i,2} + \beta_3 X_{i,3} = \boldsymbol{X\beta}$$

where $X_{i,j}$ are the features of the geographic region that center $i$ serves, and $\boldsymbol{\beta} = (\beta_0, \beta_1, \beta_2, \beta_3)^T$ is the vector of coefficients that needs to be computed. In particular, the features influencing demand are:

- $X_{i,1}$: Population of the geographic region of center $i$ (in thousands of people, 'k')
- $X_{i,2}$: Patient Visits in center $i$ (in hundreds of visits, 'h')
- $X_{i,3}$: Severity Index of cases in center $i$ (it is a rating from 1 to 10)

The company seeks a single (robust) coefficient vector ($\boldsymbol{\beta}$) that minimizes the combined prediction error across the three past operational periods ($t = 1, 2, 3$).

Table 3 contains the data for the three key features for all six centers.

| **Center** $i$ | **Population (k)** $X_{i,1}$ | **Visits (h)** $X_{i,2}$ | **Severity Index** $X_{i,3}$ |
|---|---|---|---|
| 1 | 3.0 | 12 | 7.5 |
| 2 | 2.0 | 8 | 5.0 |
| 3 | 1.5 | 6 | 4.0 |
| 4 | 4.0 | 18 | 9.0 |
| 5 | 1.0 | 4 | 2.5 |
| 6 | 2.5 | 10 | 7.0 |

**Table 3**

Furthermore, Table 4 provides the historical observed demand ($\boldsymbol{Y}^{(t)}, t = 1, 2, 3$), corresponding to the $d_i$ values in Problem 1, for each center across the three periods.

| **Center** $i$ | **Demand Period 1** $Y_i^{(1)}$ | **Demand Period 2** $Y_i^{(2)}$ | **Demand Period P3** $Y_i^{(3)}$ |
|---|---|---|---|
| 1 | 15 | 16 | 13 |
| 2 | 10 | 9 | 12 |
| 3 | 10 | 8 | 10 |
| 4 | 20 | 18 | 21 |
| 5 | 5 | 7 | 6 |
| 6 | 15 | 16 | 15 |

**Table 4**

(a) (5 points) Formulate the Multi-Objective Least Squares problem in terms of the variable vector $\boldsymbol{\beta}$ (i.e., the coefficients of the linear model). This problem minimizes the combined sum of the squared prediction errors across all three periods.

(b) (5 points) Define the normal equations of the optimization problem in part (a)

(c) (10 points) Use Python to solve the normal equations (you can use `numpy` and `np.linalg.solve`).

   i. Report the optimal coefficient vector $\boldsymbol{\beta}^* = [\beta_0, \beta_1, \beta_2, \beta_3]^T$.

   ii. Report the minimum combined Sum of Squared Errors (SSE).

(d) (15 points) The company received the latest demographic and health data, shown in Table 5. This data represents the expected conditions for the upcoming operational period. For example, the increase in Population, Patient Visits, and Severity Index for Center 1 indicates an expected rise in medical need in that region, making a prediction of required demand ($\boldsymbol{d}$) necessary.

| **Center** $i$ | **Population (k)** $X_{i,1}$ | **Visits (h)** $X_{i,2}$ | **Severity Index** $X_{i,3}$ |
|---|---|---|---|
| 1 | 3.5 | 14 | 8.0 |
| 2 | 2.0 | 9 | 5.5 |
| 3 | 1.0 | 5 | 4.5 |
| 4 | 4.5 | 19 | 9.5 |
| 5 | 1.5 | 6 | 3.0 |
| 6 | 3.0 | 11 | 7.5 |

**Table 5**

i. Use your calculated $\boldsymbol{\beta}^*$ from part (c) and the new feature data (Table 5) to predict the new raw demand vector $\boldsymbol{d}^{\text{pred}} = \boldsymbol{X}_{\text{new}}\boldsymbol{\beta}^*$. Round the predicted demands to the nearest integer.

ii. Calculate the total predicted deficit and modify the supply vector $\boldsymbol{b}$. Show your work in the following four steps:

1. Calculate the total predicted demand $D^{\text{pred}}$.
2. Calculate the total deficit by subtracting the original total supply ($\sum b_i = 75$)

$$\Delta = D^{\text{pred}} - 75.$$

iii. The company will buy the total deficit but it can only assign it to a single center. For example, if the company assigns the deficit to center 6, the modified supply vector would be

$$\tilde{\boldsymbol{b}}^{(\boldsymbol{6})} = [10, 20, 5, 15, 10, 15 + \Delta]^T$$

and the net flow requirement vector would be $\boldsymbol{R}^{(6)} = \tilde{\boldsymbol{b}}^{(\boldsymbol{6})} - \boldsymbol{d}^{\text{pred}}$. Similarly for any other center $i = 1, 2, \ldots, 6$. Determine which center is more economical to receive the entire deficit (that is, the center that will result in the lowest total transportation cost). You can use information and results from both Problem 1 and Problem 2. You must fully justify your answer.

3. *Problem 3 (35 points). Production choices*

After several years of hard work and diligent planning the R&D department of the Optimal Products Inc. has developed three (3) new products that will change the future of the world. To avoid unnecessary diversification and cannibalization of the company's existing product line, the CEO has imposed the following restrictions.

- At most two (2) out of the three new products should be produced.
- Although all three new products can be produced in the three (3) factories the company owns, just one of these factories must be selected to produce the chosen new products.
- Products can only be manufactured in the chosen factories.

The number of hours of production time needed per unit of each product and the total available time in each factory are given in Table 6:

| | Product 1 (hours/unit) | Product 2 (hours/unit) | Product 3 (hours/unit) | Available time (hours) |
|---|---|---|---|---|
| Factory 1 | 3 | 4 | 2 | 30 |
| Factory 2 | 4 | 6 | 2 | 40 |
| Factory 3 | 2 | 5 | 3 | 35 |

**Table 6**

The profit generated by selling one unit of each product and the maximum number of units the company is planning to manufacture are shown in Table 7:

| | Product 1 | Product 2 | Product 3 |
|---|---|---|---|
| Profit per unit ($) | 5 | 7 | 6 |
| Max. # units | 4 | 6 | 2 |

**Table 7**

The company wants to maximize its profit and at the same time satisfy all the restrictions posed to its production.

(a) (10 points) Formulate this problem as an integer optimization problem. Clearly define the decision variables, objective function and constraints.

(b) (10 points) Use `CVXPY` to solve this integer optimization problem. Write down the optimal solution and the optimal objective value ($Z_1^*$).

(c) (5 points) The CEO is considering relaxing the requirement "at most two products" to "all three products could be produced".

i. modify the integer optimization model in part (a) to reflect the new requirement.

ii. Use `CVXPY` to solve the new optimiztion problem.

iii. Write down the new optimal solution and maximum profit ($Z_2^*$).

iv. Calculate $Z_1^* - Z_2^*$ and explain what it means for the company.

(d) (10 points) After consultations with the strategic planning department, the CEO ask the R&D department to include the following requirement: "if product 2 is selected then product 3 must also be selected for production".

i. Formulate the CEO's requirement as a conditional constraint and add it to the integer problem in Part (a).

ii. Use `CVXPY` to find the optimal solution of the new problem.

iii. Write down the selected factory or factories, the optimal product quantities, and the final maximum profit ($Z_3^*$).

iv. Calculate $Z_1^* - Z_3^*$ and explain what it means for the company.